

AWS & DevOps – Introduction

This teaching document is intended for Day 1 classes to introduce AWS and DevOps concepts using simple explanations and visuals.

1. What is Cloud Computing?

Cloud computing is the delivery of computing resources such as servers and storage over the internet. Instead of purchasing hardware, organizations rent resources and pay only for what they use. Cloud computing represents a paradigm shift in how computing resources are acquired and managed. Instead of the traditional model where organizations had to purchase, install, and maintain their own physical data centers (servers, networking gear, cooling, power), cloud computing delivers these computing services—such as servers, storage, databases, networking, software, analytics, and intelligence—**over the internet** (the "Cloud").**The Fundamental Shift: From CAPEX to OPEX**

- **Traditional IT (CAPEX - Capital Expenditure):** Organizations made large, upfront investments (CAPEX) to buy hardware. This often led to either over-provisioning (wasted resources) or under-provisioning (inability to handle peak demand).
- **Cloud Computing (OPEX - Operational Expenditure):** Organizations *rent* these resources on a utility basis. This model allows them to treat IT costs as an operating expense (OPEX), paying only for the exact amount of resources they consume, similar to an electricity bill.

Key takeaway: Cloud computing is the practice of **renting scalable IT resources on demand** instead of the costly and restrictive process of owning and maintaining them.



2. What is AWS?

Amazon Web Services (AWS) is the world's most comprehensive and widely adopted cloud platform, offering over 200 fully featured services from data centers globally. AWS is a subsidiary of Amazon that provides on-demand cloud computing platforms and APIs to individuals, companies, and governments.**Core AWS Offerings**

AWS provides the essential building blocks for modern applications:

- **Compute:** Services like **Amazon EC2** (Elastic Compute Cloud) provide virtual servers (instances) in the cloud.
- **Storage:** Services like **Amazon S3** (Simple Storage Service) offer highly durable and scalable object storage for data, and **Amazon EBS** (Elastic Block Store) provides persistent block storage for EC2 instances.
- **Networking:** Tools like **Amazon VPC** (Virtual Private Cloud) allow users to provision a logically isolated section of the AWS Cloud where they can launch AWS resources in a virtual network they define.
- **Databases:** Managed database services such as **Amazon RDS** (Relational Database Service) and **Amazon DynamoDB** (NoSQL database) simplify setup and maintenance.

Why AWS is Dominant: AWS revolutionized the market by providing a highly **scalable**, **secure**, and **cost-effective** environment, freeing companies from the burden of physical infrastructure management so they can focus on their core business innovation.

AWS Global Infrastructure: Regions and Availability Zones

AWS's global infrastructure is the physical foundation that makes cloud computing possible.



AWS Regions Overview

AWS infrastructure is divided into **Regions**, each representing a distinct geographical location (e.g., North Virginia, Mumbai). This division is fundamental for high availability, disaster recovery, and compliance.

- Every region has a unique, short code (e.g., `us-east-1`, `ap-south-1`).
- When provisioning resources (like EC2 instances or S3 buckets), you must select the region where they will reside.
- Resources are **region-specific** and generally do not cross regions unless explicitly configured (e.g., cross-region replication).

Common AWS Regions and Codes

Geography	Region Name	Code	Note for Students
United States	N. Virginia	`us-east-1`	**Most commonly used** (often the default).
	Ohio	`us-east-2`	
	N. California	`us-west-1`	
	Oregon	`us-west-2`	
Asia Pacific	Mumbai (India)	`ap-south-1`	Choose closest to your target users.
	Singapore	`ap-southeast-1`	
	Sydney	`ap-southeast-2`	
	Tokyo	`ap-northeast-1`	
	Seoul	`ap-northeast-2`	
Europe	Frankfurt	`eu-central-1`	
	Ireland	`eu-west-1`	
	London	`eu-west-2`	
Canada	Central	`ca-central-1`	
South America	São Paulo	`sa-east-1`	

Key Decision Factors for Region Selection:

1. **Latency:** Select the region closest to your end-users to minimize response time.
2. **Compliance:** Choose regions that meet specific regulatory requirements (e.g., data residency laws in Europe or India).
3. **Cost and Service Availability:** Pricing and the availability of certain specialized services can vary between regions.

Availability Zones (AZs)

Each AWS Region is composed of multiple, isolated locations known as **Availability Zones (AZs)**.

- An AZ is one or more discrete data centers, each with redundant power, networking, and connectivity.
- They are physically separated from each other by meaningful distance (typically miles) to prevent a single disaster (e.g., a flood or fire) from affecting multiple AZs, yet they are close enough for low-latency network connectivity.
- The use of multiple AZs within a single region is the primary method for architecting **high availability** and **fault tolerance** in AWS.

Key takeaway: AWS is the leading cloud platform that provides the physical, secure, global infrastructure (servers, storage, networking) and the managed services built upon it.



3. What is DevOps?

DevOps is not a product or a single tool; it is a **methodology, a cultural philosophy, and a set of practices** that aims to streamline the process of software delivery and infrastructure changes. It is a portmanteau of "Development" (Dev) and "Operations" (Ops).**The DevOps Goal: Breaking Down Silos**

Traditionally, development and operations teams worked in separate silos with conflicting goals:

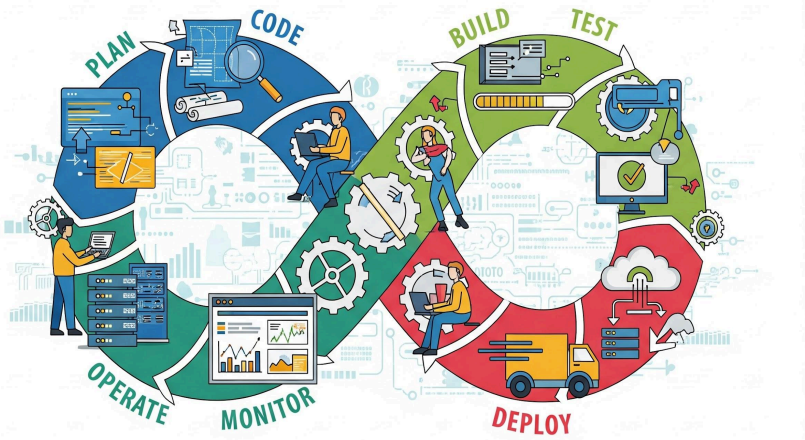
- **Development:** Focused on rapid feature releases and change.
- **Operations:** Focused on system stability and reliability (resisting change).

DevOps bridges this gap by advocating for **increased communication, collaboration, and integration** between these two groups, allowing businesses to innovate faster and more reliably.**Key Practices of the DevOps Methodology**

DevOps is implemented through the adoption of specific practices, often referred to as the "CAMS" or "CALMS" model:

1. **Automation:** Automating repetitive tasks across the entire lifecycle (testing, building, deployment, infrastructure provisioning).
2. **Continuous Integration (CI):** Developers merge code changes into a central repository frequently, where automated builds and tests are run.
3. **Continuous Delivery/Deployment (CD):** Once code passes CI, Continuous Delivery ensures it can be released to a production-like environment automatically, while Continuous Deployment automatically pushes every change that passes all gates to production.
4. **Monitoring & Feedback:** Implementing tools to continuously monitor application performance and infrastructure health in real-time, providing immediate feedback to both Dev and Ops teams.
5. **Culture:** The most critical component—a culture that fosters sharing, mutual responsibility, and failure tolerance.

Key takeaway: DevOps is the process, culture, and toolchain focused on automation and collaboration to build, test, and deliver applications quickly, reliably, and consistently.-----**Comprehensive Summary and Interrelation**



Summary

Concept	What it is	Role
AWS	A cloud platform and set of services	Provides the infrastructure
DevOps	A set of practices and culture	Provides the process to utilize the infrastructure efficiently

AWS provides the cloud infrastructure, while DevOps provides the process and practices to build, deploy, and monitor applications efficiently.**AWS & DevOps – A Comprehensive Introduction.**

The relationship between Cloud Computing, AWS, and DevOps is synergistic: AWS provides the technological foundation, and DevOps provides the methodology to leverage that foundation efficiently.

Concept	What it is	Primary Role	Value Proposition
Cloud Computing	The delivery model of IT resources over the internet.	The overarching concept of **renting** IT.	Eliminates the need for costly physical infrastructure

			ownership.
AWS	A robust platform of cloud services and global infrastructure.	Provides the **infrastructure and technological tools** (IaaS, PaaS, SaaS).	Offers elasticity, security, and global scale on-demand.
DevOps	A cultural, process, and technical methodology.	Provides the **process and best practices** for utilizing the infrastructure.	Ensures rapid, reliable, and automated delivery of software onto the cloud platform.

In essence: AWS provides the scalable cloud infrastructure, while DevOps provides the process, automation, and culture necessary to build, deploy, and monitor applications **efficiently and rapidly** on the AWS platform.